CYFERD SPORTS

Modern Football
Sports Science App

Objective and Context

All football clubs have a sports science structure, although its form may vary.

Some appoint a dedicated Performance or Sports Science Director, while others rely on the First Team Fitness Coach. Regardless of structure, the objective is always the same:

To keep players fit, available, and performing at the highest level—while preventing injuries and accelerating recovery.

This project proposes a **single integrated platform** that connects all health and performance layers, automates data processing, and transforms information into **clear, actionable insights for decision-makers.**

Core Objectives

- Prevent injuries using **real-time and historical data patterns**
- Keep players **fit, available, and stronger than before** - Availability is the most valuable "performance KPI"
- Create a **single source of truth** across all departments
- Turn data into **clear decisions**: training load, return-to-play, selection, and contract/value

Worflow

Collect Data

From different providers.

Data that sometimes is forgotten.

Can be from national team, collective and even personal like Whoop!

Automate Procesing

Daily or weekly pipelines

A lot of files are in pdf shared by pdfs and whatsapp shared

Analyze

Find patters, correlations with historical years

Accountability

As time goes with this solutions the model learn from all the data inputs and is more intelligent

Accountability because with this app you can do what you say you're going to do in terms of players projection

Predict risk

Probability of success against historical cases, or other ideal projects

Communicate on time

Alerts, tasks, calendar, reports for the decition makers on time

Pilars of Sports Science App

Communication & Workflow Automation

(tasks, calendar, messaging, alerts)

Performance & Load Monitoring

GPS match/training intensity

Key for players availability and predict Load for upcoming matches

Nutrition & Hydration

anthropometrics, compliance, standards

Fundamental for pre season to set parameters are key, also for the regulations in the squad (weighting)

Medical & Injury Prevention

Historical Data, Injury paterns, diagnosis, Return to play Program

Historical Data can show patterns and yellow flags of proor cases

Testing & Wellness

objective tests, subjective questionnaires

Red flags can signal that an injury is coming, and detecting a flu early can help prevent it from spreading through the whole squad.

Multiple Data Sources

Because Data comes from different sources is important that is standardized and that the client chose a certain perferable source as some as similar parameters,

Standardize the Data

Layers

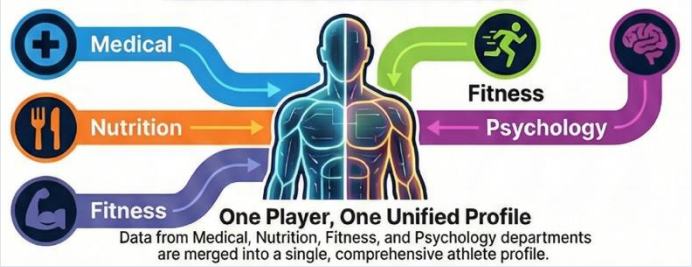
- Player ID + timelines (training, match, RTP, medical events)

Standard metrics:

- Injuries (type, location, mechanism, severity, days lost)
- Load definitions (HSR zones, sprint thresholds, weekly totals)
- Return to play stages (phase 1–X, criteria to progress)
- Nutrition “in range / out of range” standards

Data quality rule example:

- Missing data alerts, outlier checks, auto-cleaning



Example of Data Sets

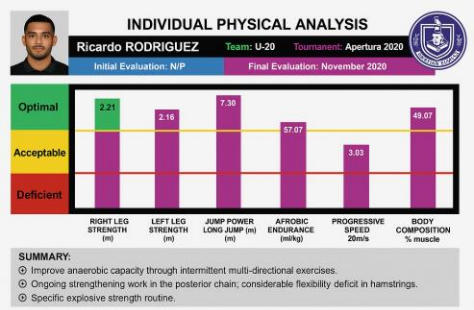
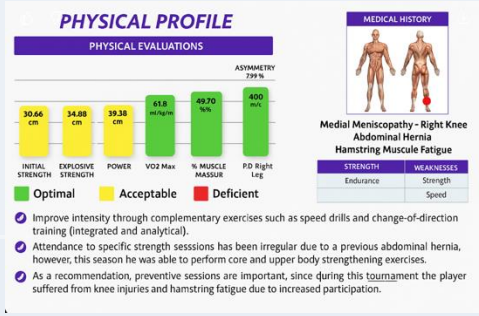
- GPS / Wearables: distance, HSR, sprints, accelerations, decels, session load.
- Match Data / Tracking: match demands, intensity trends, minutes, peaks
- Medical: injuries, diagnoses, treatment plan, RTP stage, availability status
- Wellness: sleep, fatigue, soreness, stress, readiness self-report
- Testing: speed, endurance, strength, asymmetry tests, benchmarks
- Nutrition: weight, fat mass, muscle mass, hydration / compliance flags
- Thermography (if club has cameras).

Today’s reality

Data lives in silos: GPS / medical notes / nutrition files / wellness forms / imaging

Manual reporting, slow follow-up, inconsistent standards

Decision-makers get “status updates” instead of “risk + reason + recommendation”

Desired

1. **Integration + automation + shared language across staff**
2. **One dashboard for: risk, readiness, return-to-play stage, and availability**